CHE213 Innovation Presentation

Mass Transfer Study of Solid-Liquid Dissolution

Group-19
Hetal Sharad Patil 240449
Paridhi Rastogi 240731
Swati 241077
T Krishnapriya Reddy 241080

Objectives:

- To calculate the mass transfer coefficient(kL) for the benzoic acid–water system
- To determine the effective stagnant film thickness(delta) using film theory
- To evaluate the system using dimensionless parameters: Sherwood number(Sh) and Schmidt number(Sc)

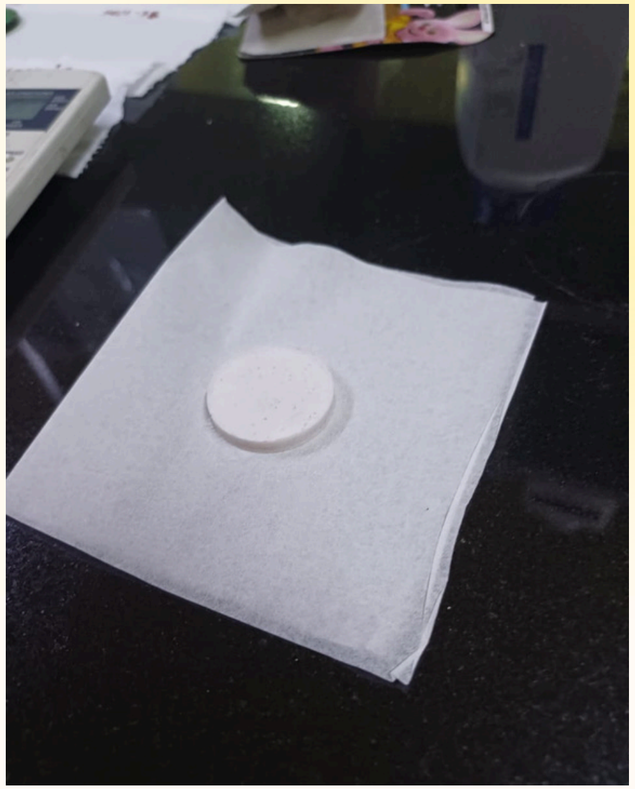
Materials:

Benzoic Acid Pellet: Diameter = 32 mm, Thickness = 5.65 mm, Mass = 5.473 g
Solvent: 1000 mL distilled water
Titrant: 0.02 N sodium hydroxide (NaOH) solution
Indicator: Phenolphthalein
Apparatus: Magnetic stirrer, beaker (1000 mL), burette, 10 mL pipette

Experimental setup:

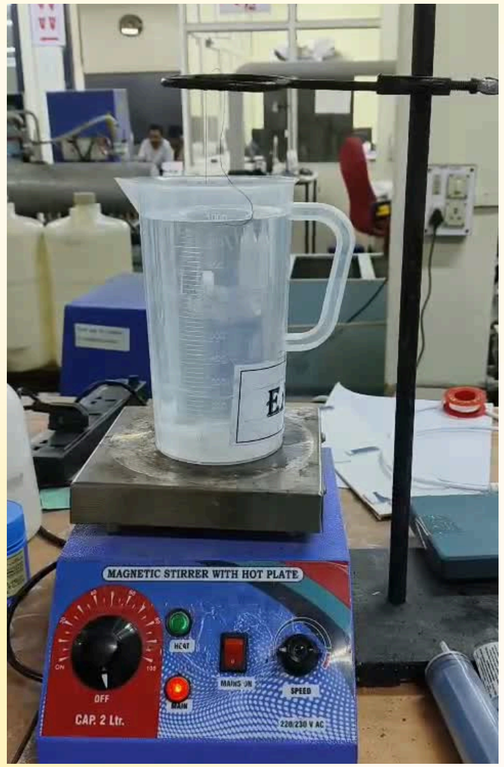


Hydraulic press



Benzoic acid pellet

Experimental setup:



Magnetic stirrer in action



Titrated sample



Complete setup

Methodology:

1. Measure the pellet dimensions using vernier calipers
2. Fill beaker with 1000 mL distilled water
3. Turn on the magnetic stirrer
4. Introduce the benzoic acid pellet at t=0
5. Withdraw 10 mL samples at regular intervals
6. Prepare 0.02 N NaOH from NaOH pellets
7. Titrate each sample using 0.02 N NaOH
8. Continue this process until saturation(~64 minutes)

Theory

Rate of Dissolution

$$\frac{dC}{dt} = k_L \left(\frac{A}{V}\right)(C_s - C)$$

Linearized Form

$$\ln\left(\frac{C_s}{C_s - C}\right) = \left(\frac{k_L A}{V}\right) t$$

$$\text{Slope } (S) = -\frac{k_L A}{V}$$

Film Thickness

$$\delta = \frac{D_{AB}}{k_L}$$

Mass Transfer Coefficient

$$k_L = S \cdot \frac{V}{A}$$

Predictive Correlation

$$Sh = a\, Re^n\, Sc^m$$

Sherwood Number

$$Sh = \frac{k_L\, d}{D_{AB}}$$

Schmidt Number

$$Sc = \frac{\nu}{D_{AB}}$$

Reynolds Number

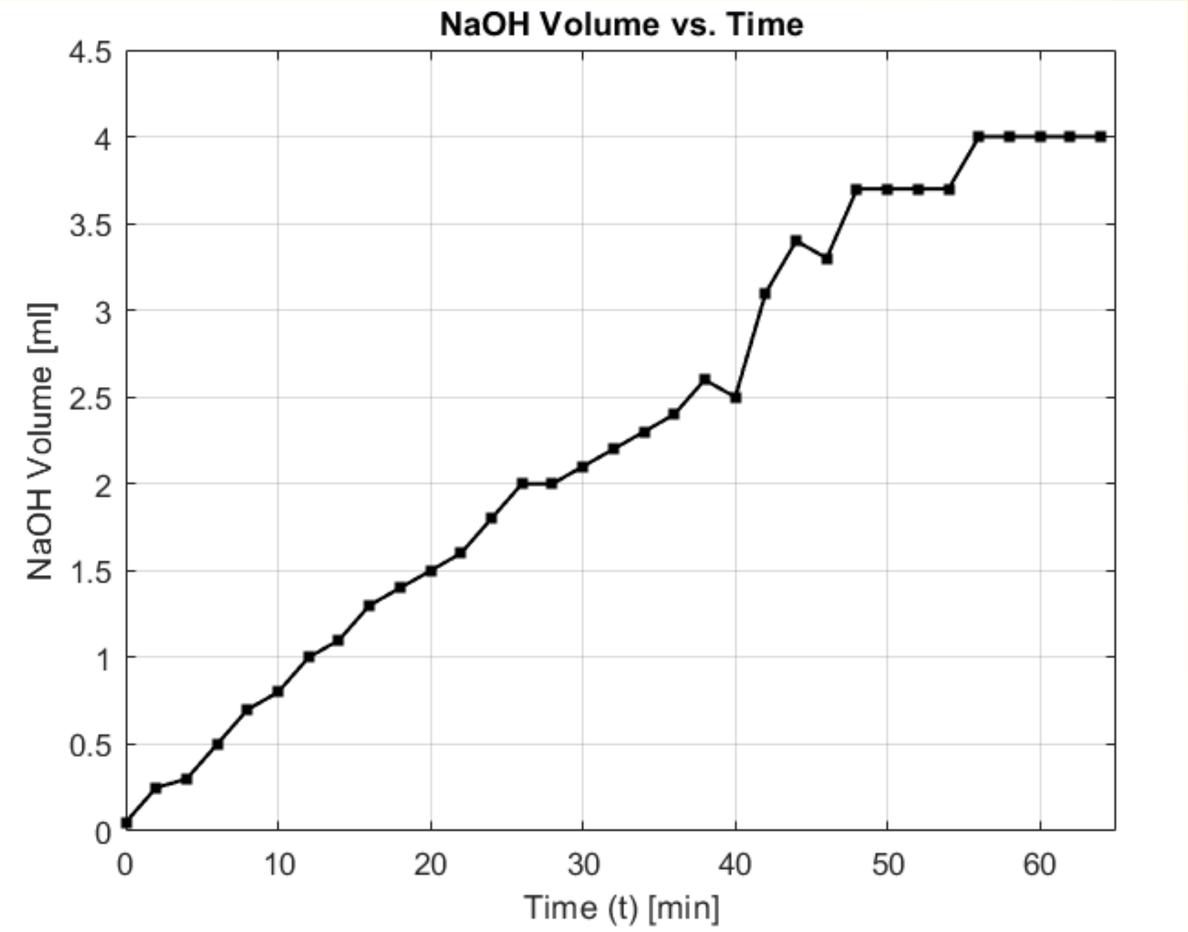
$$Re = \frac{N\, d^2}{\nu}$$

Raw Data

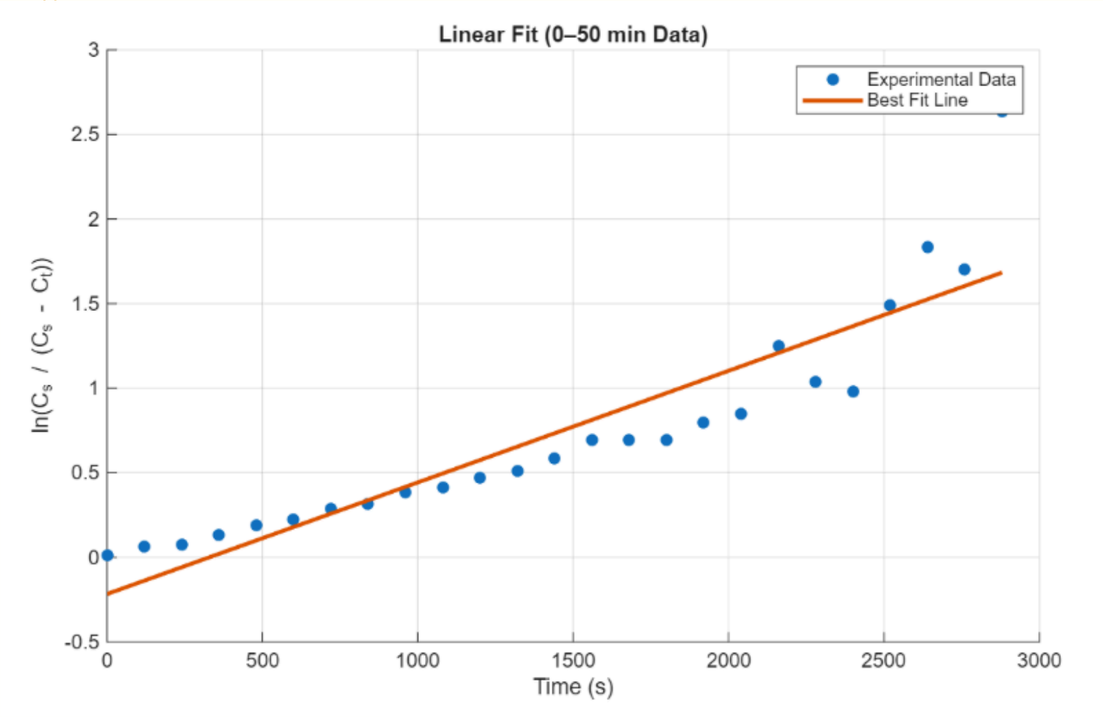
Time (t) (min)	NaOH volume (ml)
0	0.05
2	0.25
4	0.3
6	0.5
8	0.7
10	0.8
12	1
14	1.1
16	1.3
18	1.4
20	1.5
22	1.6
24	1.8
26	2
28	2
30	2.1
32	2.2
34	2.3

Time (t) (min)	NaOH volume (ml)
36	2.4
38	2.6
40	2.5
42	3.1
44	3.4
46	3.3
48	3.7
50	3.7
52	3.7
54	3.7
56	4
58	4
60	4
62	4
64	4

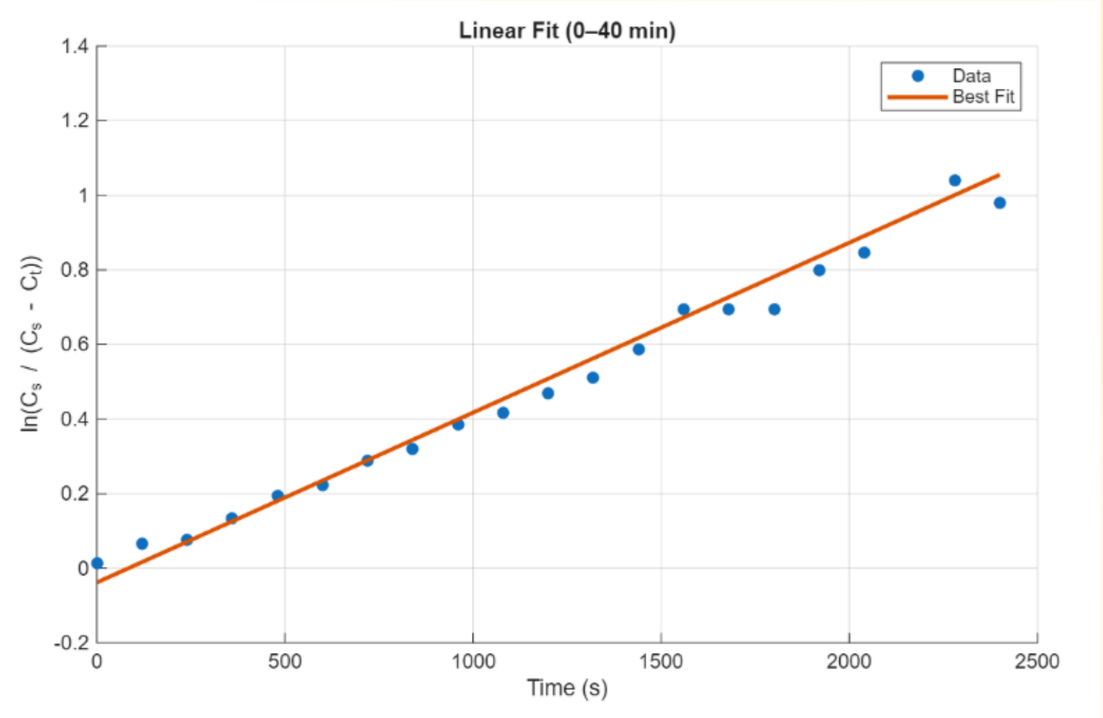
NaOH volume v/s time:



kL calculation:



Slope (S) = 6.5977e-04 s^-1
R^2 value = 0.8272



Slope (S) = 4.5558e-04 s^-1
R^2 = 0.9478

Extending the dataset to 50 minutes resulted in a higher slope but a significantly lower R^2 value, indicating deviation from linearity due to saturation effects. Hence, the linear region (0–40 min) was used for accurate estimation of kL.

Thus we took S = 4.55 * 10^(-04) s^-1

Code for data fitting:

C:\Users\Hetal\Downloads\Hetal_240449_aa2c9235-2b6b-4ae9-84f0-960d510231ab\untitled7.m

```
clc; clear; close all;

% Time (min)
t_min = [0 2 4 6 8 10 12 14 16 18 20 22 24 26 28 30 32 34 36 38 40];

% Convert to seconds
t = t_min * 60;

% ln(Cs/(Cs - Ct))
ln_vals = [0.0126 0.0645 0.0770 0.1335 0.1931 0.2231 0.2877 0.3185 ...
           0.3840 0.4155 0.4700 0.5108 0.5878 0.6931 0.6931 0.6931 ...
           0.7990 0.8473 1.2528 1.0397 0.9808];

% Linear regression
p = polyfit(t, ln_vals, 1);
slope = p(1);
intercept = p(2);

% Predicted values
ln_pred = polyval(p, t);

% R^2 calculation
SS_res = sum((ln_vals - ln_pred).^2);
SS_tot = sum((ln_vals - mean(ln_vals)).^2);
R2 = 1 - (SS_res / SS_tot);
```

```
% Plot
figure;
scatter(t, ln_vals, 'filled'); hold on;
plot(t, ln_pred, 'LineWidth', 2);

xlabel('Time (s)');
ylabel('ln(C_s / (C_s - C_t))');
title('Linear Fit (0-40 min)');
legend('Data', 'Best Fit');
grid on;

% Output
fprintf('Slope (S) = %.4e s^-1\n', slope);
fprintf('R^2 = %.4f\n', R2);
```

Data Analysis:

Pellet dimensions:

r = 0.016 m h = 0.00565 m

A = 2*pi* r2 + 2*pi* r* h A = 0.002176 m2

Linearized equation:

ln (Cs/ (Cs - Ct)) = (k_L*A/V) *t

S = k_L*A/V

$$k_L = \frac{S \cdot V}{A}$$

Taking V=1000 mL

kL=2.09x10^(-4)

Taking average volume=(1000+660)/2=830 ml

(660 is the volume left out)

kL=1.735x10^(-4)

According to literature, kL comes out to be 2.7×10^{-5} to 1.5×10^{-4} m/s

Reference: https://pubs.acs.org/doi/10.1021/acs.iecr.7b04730

Data Analysis:

For benzoic acid in water:

At 25°C:

DAB ≈1.0×10−9m2/s

At 28°C:

DAB ≈1.08×10−9m2/s

(DAB∝T/μ)

Film Thickness:

delta = D_AB / k_L

delta = (1.08*10^-9) / (1.735*10^-4)

delta = 6.22*10^-6 m = 6.22 micron

Predictive Correlation

$$\frac{Sh}{Sc^{1/3}} = \frac{5140}{9.16} \approx 561.13$$

$$Sh = 5.49\, Re^{0.67}\, Sc^{1/3}$$

Dimensionless parameters:

Schmidt number:

Kinematic viscosity: v = 8.3×10^−7 m²/s

D_AB = 1.08 × 10⁻⁹ m²/s

Sc=ν/D_AB=(8.3x10^-7)/(1.08x10^-9)=768.51

Sherwood number:

Sh=k_L×d/D_AB

Sh=(1.735×10⁻⁴)×(0.032)/(1.08×10⁻⁹)

Sh=5.552×10⁻⁶/1.08×10⁻⁹

Sh≈5140

Results and Conclusion:

We successfully computed the values of kL for forced benzoic acid dissolution in water at 28 degrees celcius
We also found out the film thickness from film theory and computed dimensionless parameters i.e., Schmidt and Sherwood Numbers

The results are as follows:

$kL = 1.735 \times 10-4 m/s$

$\delta = 6.22 \mu m$

$Sc = 768.51$

$Sh = 5140$

Forced convection significantly reduces boundary layer thickness
Mass transfer is strongly enhanced by stirring
The system follows film theory in the initial linear regime

Overall, the study provides important insight into dissolution kinetics and highlights the role of hydrodynamics in controlling mass transfer rates, which is critical for industrial processes such as extraction, crystallization, and reactor design.

Results and Conclusion:

We also found out the **fitted correlation** between Sh, Sc, Re.

Predictive correlation:

$Sh = 5.48 \times Re^{0.67} \times Sc^{(1/3)}$

Sc ≈768

Sh ≈5140

These values indicate:

High Sc: mass diffusion is much slower than momentum diffusion

High Sh: strong enhancement of mass transfer due to convection

Thus, the experiment validates classical mass transfer theory under forced convection conditions.

Precautions:

- Ensure the stirring speed remains constant throughout the experiment — any variation in speed changes h and k_L, making the linearized plot non-linear
- Withdraw samples quickly and titrate promptly to prevent continued dissolution in the sample after withdrawal
- Ensure the phenolphthalein endpoint is a stable pale pink, not a deep pink — overshooting the endpoint overestimates concentration
- Do not disturb the pellet or beaker during sampling — mechanical disturbance changes the local flow field and produces inconsistent k_L values
- Record ambient temperature before and after the experiment — benzoic acid solubility is temperature sensitive and affects C_s
- Account for volume depletion due to repeated 10 mL sampling — for a 1000 mL system, after 10 samples this is a 10% volume reduction which may affect late-time concentration calculations
- Verify NaOH normality before the experiment using a primary standard such as oxalic acid

Improvements:

- Repeat the experiment at multiple stirring speeds (e.g., 100, 200, 400, 600 RPM) — this generates multiple (Re, Sh) data pairs, allowing independent determination of both a and n in the Sh = a·Re^n·Sc^m correlation
- Control water temperature precisely at 25°C using a water bath — this allows direct comparison of C_s with the known theoretical value and removes temperature as an unknown variable
- Use pellets of different sizes (varying d) to vary Re independently of stirring speed, providing an additional axis for correlation fitting

Thank You

We would like to express our sincere gratitude to our course instructor for their valuable guidance, support, and encouragement throughout this project.
We also extend our heartfelt thanks to the laboratory staff for their assistance in setting up the experimental apparatus and for providing the necessary resources to carry out this work smoothly.